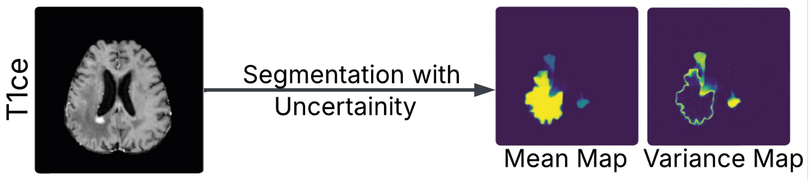
Interactive Lung Tumor Segmentation using Denoising Diffusion Models

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Memorial Sloan Kettering Cancer Center

INTRODUCTION

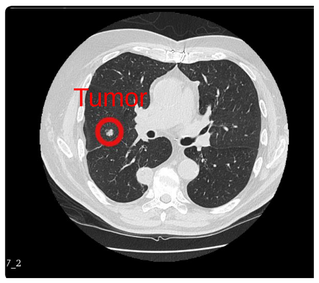
Denoising Diffusion Model (DDM) for Tumor Segmentation:



Challenges with DDMs:

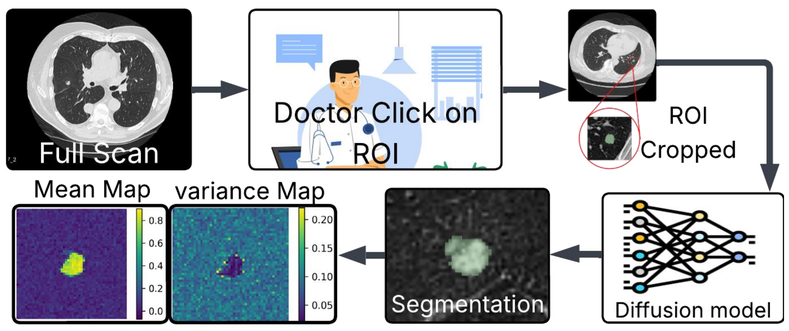


Slow DDPM 3D, Inference ~Hours, Training ~ weeks



Accuracy: Hard to segment small structures

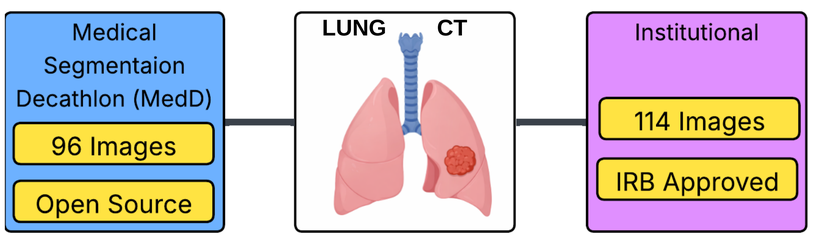
Human Assisted DDM (HA-DDM):



Materials and Methods

Datasets:

3D Dataset:



2D Dataset:

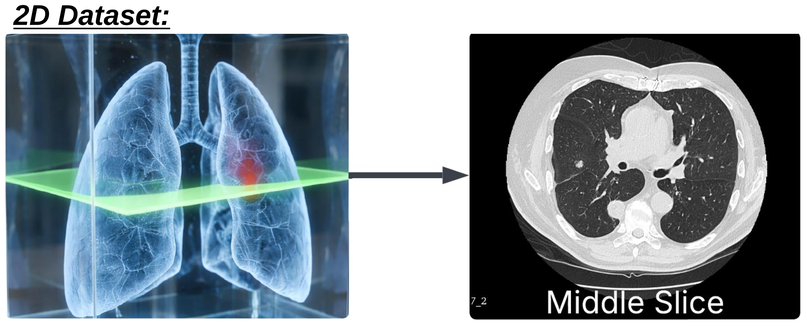


Figure 1. 2D Dataset created from 3D scans

Results

2D

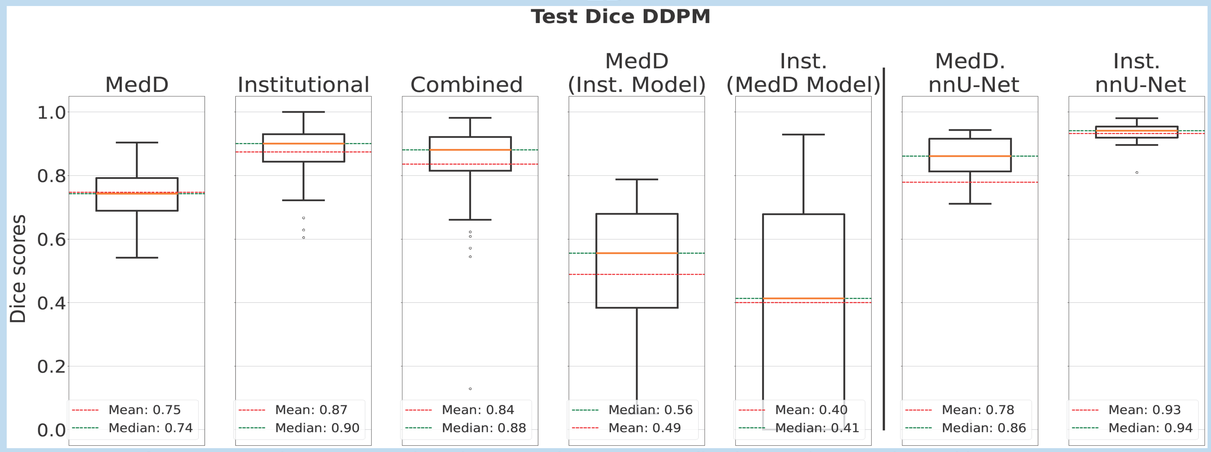


Figure 2. 2D Dice score: In-domain, cross-dataset generalization, and comparison with nnU-Net.

3D

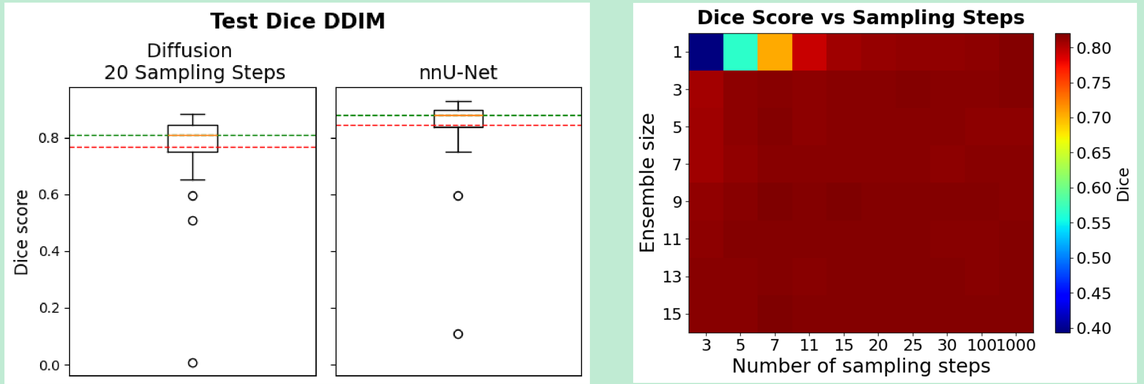


Figure 3. LEFT: 3D Test data dice scores: Diffusion vs nnU-Net. RIGHT: Dice for different ensemble size and sampling steps

HA-DDM VS Full Image Runtime

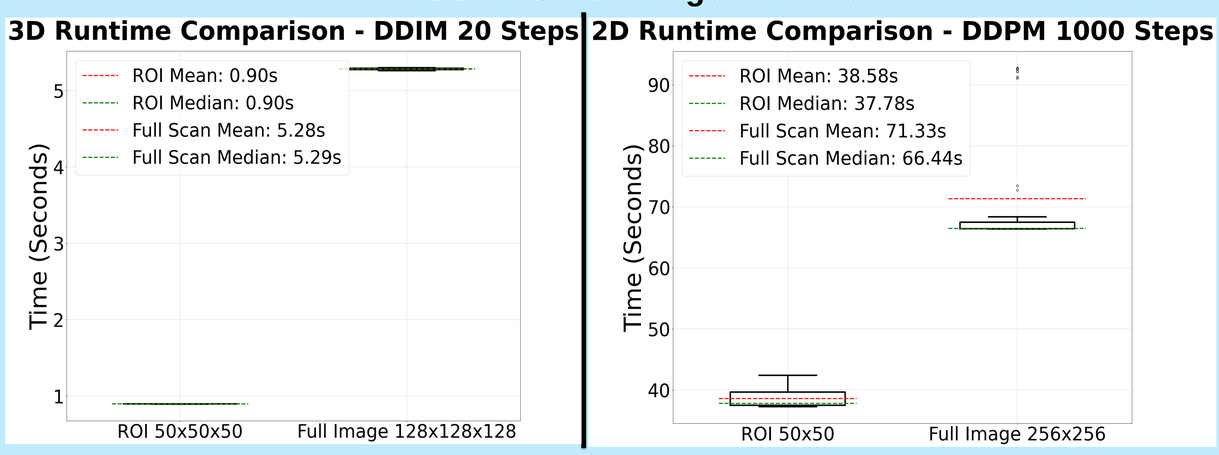


Figure 4. Inference runtime of HA-DDM ROI vs Full Image in 3D and 2D

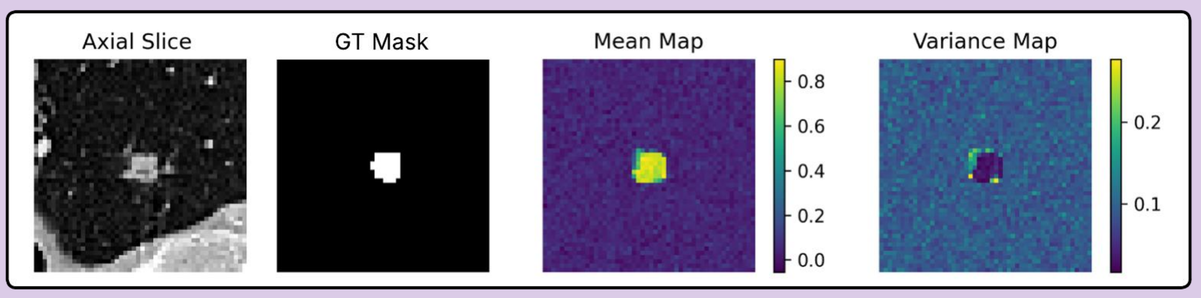
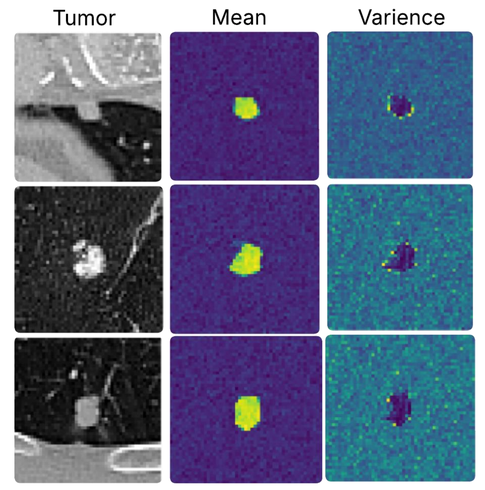


Figure 5. Diffusion-based segmentation: mean and variance uncertainty around tumor boundaries.



Conclusion

- **HA-DDM** performs well in 2D and 3D; results comparable to nnU-Net.
- Cross-dataset **generalizability** remains challenging (MedD vs Institutional)
- Ensemble sampling **(5)** with fewer denoising sampling steps **(20)** preserves performance reducing runtime significantly
- Variance maps from diffusion models **reveal uncertainty** not captured by deterministic segmentation.

References

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[5] H. Wang, Z. Liu, K. Sun, X. Wang, D. Shen, and Z. Cui, "3D MedDiffusion: A 3D Medical Latent Diffusion Model for Controllable and High-quality Medical Image Generation," *IEEE Trans Med Imaging*, p. 1, 2025, doi: 10.1109/TMI.2025.3585372.